higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

16 February 2022 (X-paper)
09:00–12:00

Calculators may NOT be used,
but drawing instruments may be used.

This question paper consists of 6 pages.

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282Q1E2216

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a new page.
5. Use only a black or blue pen.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 $10_8 + 10_{10} =$

A 18
B 20
C 28
D 30

1.2 $24_{16} – 24 =$

A 20_{16}
B 10_8
C 20_8
D 1100_2

1.3 $10{,}01_2 – 2_8 =$

A 0,5
B 0,25
C 0,75
D 1

1.4 $4_8 \times 4_{16} =$

A 10_{16}
B 6_8
C 16_{16}
D 8_{16}

1.5 $20_8 \div 10_2 =$

A 0
B 2
C 4
D 8

(5 × 2) **[10]**

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QUESTION 2

Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (2.1–2.8) in the ANSWER BOOK.

2.1 $A + A + 0 = 1$

2.2 $A + \overline{A} + 1 = 0$

2.3 $A + \overline{A} + 0 = 1$

2.4 $A \bullet A \bullet 1 = A$

2.5 $A \bullet \overline{A} \bullet 1 = 1$

2.6 $A \bullet \overline{A} \bullet 1 = 1$

2.7 $A + (A \bullet \overline{B}) = A$

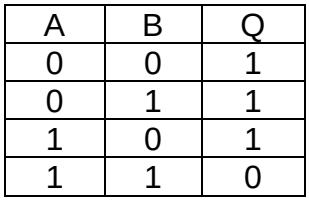
2.8 $\overline{A} \bullet (\overline{A} + \overline{B}) = \overline{A}$

(8 × 1) **[8]**

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QUESTION 3

Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–G) next to the question number (3.1–3.7) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1	Logic gate that provides a binary 1 on the output when all inputs are binary 1s	A	parallel
3.2	Weighed self-inverting code	B	Johnson
3.3	Twisted ring counter	C	AND gate
3.4	Maximum count for a 4-bit asynchronous binary counter	D	NAND gate
3.5	 Truth table	E	trailing
3.6	Edge where a master-slave JK flip-flop will change the output data	F	1111_2
3.7	Type of output data provided by a demultiplexer	G	2421 BCD

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(7 × 1) **[7]**

TOTAL SECTION A: 25

SECTION B

QUESTION 4

Complete each of the following calculations in the binary number system and clearly show all steps:

4.1 $1001,01_2 - 10000,0101_2$ (Use the 1's complement method.)

4.2 $1001,01_2 - 100,011_2$ (Use the 2's complement method.)

4.3 $10010,01_2 \times 1001,11_2$

4.4 $10110001,1111_2 \div 1001,11_2$

(4 × 5) **[20]**

QUESTION 5

Draw the circuit diagrams of each of the following Boolean expressions. Do not change or simplify the expressions:

5.1 $\overline{(A \oplus \overline{B \bullet C})} \bullet \overline{(A \bullet B + \overline{A} \bullet B \bullet C)} = f$

5.2 $A \bullet (A \bullet \overline{B + C}) \oplus \overline{(A \bullet B \bullet C)} = f$

5.3 $\overline{(\overline{A + B} + \overline{C})} \bullet \overline{(\overline{A} + \overline{B} + C)} = f$

(3 × 5) **[15]**

QUESTION 6

Simplify each of the following Boolean expressions:

6.1 $\overline{(A + B\overline{C} + \overline{A}B)} + \overline{(\overline{A} + \mathrm{BC} + A\overline{B})} = f$

6.2 $\overline{(A \bullet \overline{B} + \overline{A} \bullet \overline{B} \bullet C)(\overline{A} \bullet \overline{B} \bullet C + \overline{A})}\,\overline{(\overline{A} \bullet \overline{B} \bullet C + A \bullet \overline{B})} = f$

(2 × 5) **[10]**

QUESTION 7

7.1 Draw the circuit diagram of a 4-bit parallel adder by using full adders. (10)

7.2 Differentiate between a *full adder* and a *half adder*. (2 + 2) (4)

7.3 Make a neat, labelled sketch of a 4-bit twisted ring counter that uses J-K flip-flops. (6)

7.4 Draw a neat, labelled circuit diagram of a 4:1-line multiplexer.

Show all input, output and control points clearly. (10)

[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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